

3.1 Baye's Rule

¹
Soln:- Here,

Probability of having Cancer = $P(C) = 0.02$

Probability of not having Cancer = $P(\bar{C}) = 0.98$

Probability that if someone has cancer,
the test says +ve 80% of the time

$$P(C^+) = 0.80$$

Probability that if someone doesnot have Cancer,
still the test say +ve 9.6% of the time

$$P(\bar{C}^+) = 0.096$$

If the ~~the~~ test is +ve, the probability of having
Cancer

$$P(C^*) = ?$$

$$P(C^*) = \frac{0.8 \times 0.02}{0.8 \times 0.02 + 0.096 \times 0.98}$$

$$P(C^*) \approx 0.145$$

The probability that someone has Cancer when tested
+ve is 14.5%.

2.

The quiz assumed a 20% false +ve rate.

It ~~was~~ was correct 80% of the time.

But here in this problem, the false +ve rate is 9.6%, so there are fewer false positives and the probability becomes higher.

3.2

① Find the permutation of having even +ve number (4 digits) and ② also find the permutation of having cell number.

The cell number must start with 9 and end on 1. The number must be in ten digits. ③ A website ~~has~~ has four digits pin number (alphanumeric). The pin must start with vowel and end with consonant. (Repetition is not allowed). How many

permutations are possible. ④ from the ~~group~~ people of 8, How many group of 3 is possible.

⑤ ~~From the~~ A Court has 12 people. 5 judges and 7 lawyer. How many group of people having 5 person, but the committee must have exactly 3 ~~lgy~~ lawyers. ~~are~~ are possible?

Soln:-

1

$$\underline{9} \quad \underline{10} \quad \underline{10} \quad \underline{5}$$

even $\rightarrow 0, 2, 4, 6, 8 \rightarrow$ (must end on

first place $\rightarrow 1, 2, 3, 4, 5, 6, 7, 8, 9$

$$= 9 \times 10 \times 10 \times 5 = 4500$$

2.

$$\underline{1} \quad \underline{10} \quad \underline{10} \quad \underline{10} \quad \underline{10} \quad \underline{10} \quad \underline{10} \quad \underline{10} \quad \underline{1}$$

$$= 10^8$$

3.

$$\begin{array}{ccccccc} \underline{5} & \underline{34} & \underline{33} & \underline{21} & & = & 5 \times 34 \times 33 \times 21 \\ \text{V} & & & & & & \\ \text{Vowel} & & & \text{Consonant} & = & 117,810 \end{array}$$

4.

$$\begin{aligned}
 {}^8C_3 &= \frac{8!}{(8-3)! 3!} = \frac{8!}{5! 3!} \\
 &= \frac{7 \times 8 \times \cancel{6}}{\cancel{6}} = 56
 \end{aligned}$$

5.

The group or committee has 5 people and in the committee we must have 3 lawyers. We have 7 lawyers and 5 Judges. So the req^d combination would be

$$\begin{aligned}
 {}^7C_3 \times {}^5C_2 \\
 = \frac{7!}{4! 3!} \times \frac{5!}{3! 2!}
 \end{aligned}$$

$$= \frac{7 \times \cancel{6} \times 5}{\cancel{6}} \times \frac{5 \times \cancel{4}}{\cancel{2}}$$

$$= 350$$

9.3

Soln:-

Here,

$$P(X=x) = {}^n C_x p^x q^{n-x} \quad (q=1-p)$$

$$E(h(x)) = \sum_x h(x) P(x)$$

1.

$$n=3$$

$$p = \frac{1}{2}$$

$$E(h(x)) =$$

$$P(X=x) = {}^3 C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{3-x}$$

$$E[h(x)] = \sum_x h(x) P(x)$$

$$x = 0, 1, 2, 3$$

$$P(X=0) = {}^3 C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{3-0}$$

$$= \frac{3!}{0! 0!} \times 1 \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$P(X=1) = {}^3 C_1 \left(\frac{1}{2}\right) \left(\frac{1}{2}\right)^2 = \frac{3!}{2! 1!} \left(\frac{1}{2}\right)^3 = \frac{3}{8}$$

$$P(X=2) = {}^3C_2 \left(\frac{1}{2}\right)^2 \frac{1}{2}$$

$$= \frac{3!}{2!} \frac{1}{8} = \frac{3}{8}$$

$$P(X=3) = {}^3C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^0$$

$$= \frac{3!}{0! 3!} \frac{1}{8} = \frac{1}{8}$$

$$E(h(X)) = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8}$$

$$= \frac{3}{8} + \frac{6}{8} + \frac{3}{8} = \frac{102}{8}$$

$$= \boxed{1.5}$$

2. Let us prove $E[X] = np$

We know that, $E(X) = \sum_{x=0}^n x P(X=x)$

$$E(X) = \sum_{x=0}^n x {}^nC_x p^x (1-p)^{n-x}$$

$$= \sum_{x=0}^n x \frac{n!}{(n-x)! x!} p^x (1-p)^{n-x}$$

Why infinity limit now?

$$\begin{aligned}
 &= \sum_{x=0}^{\infty} \frac{n!}{(n-x)! x (x-1)!} p^x (1-p)^{n-x} \\
 &= \sum_{x=0}^{\infty} \frac{n!}{(n-x)! (x-1)!} p^x (1-p)^{n-x} \\
 &= \sum_{x=0}^{\infty} \frac{n (n-1)!}{(n-x) (x-1)!} p^x (1-p)^{n-x} \\
 &= \sum_{x=0}^{\infty} n (n-1) C_{x-1} p^x (1-p)^{n-x}
 \end{aligned}$$

$$= np \sum_{x=1}^{\infty} (n-1) C_{x-1} p^{x-1} (1-p)^{n-x}$$

Let $k = x - 1$

What happened to infinity limit?

$$= np \sum_{x=0}^{n-1} (n-1) C_k p^k (1-p)^{(n-1)-k}$$

From binomial theorem,

$$(a+b)^n = \sum_{K=0}^n \binom{n}{K} a^K b^{n-K}$$

So, we can write,

$$\sum_{K=0}^{n-1} \binom{n-1}{K} p^K (1-p)^{n-1-K}$$

$$= (p + (1-p))^{n-1}$$

$$\text{and } p + (1-p) = 1$$

$$\text{So, } \cancel{\sum_{K=0}^{n-1}} [p + (1-p)]^{n-1} = 1^{n-1} = 1$$

$$\therefore \boxed{E(X) = np}$$

and in our case

$$E(X) = 3 \cdot \frac{1}{2} = 1.5 \text{ \#}$$